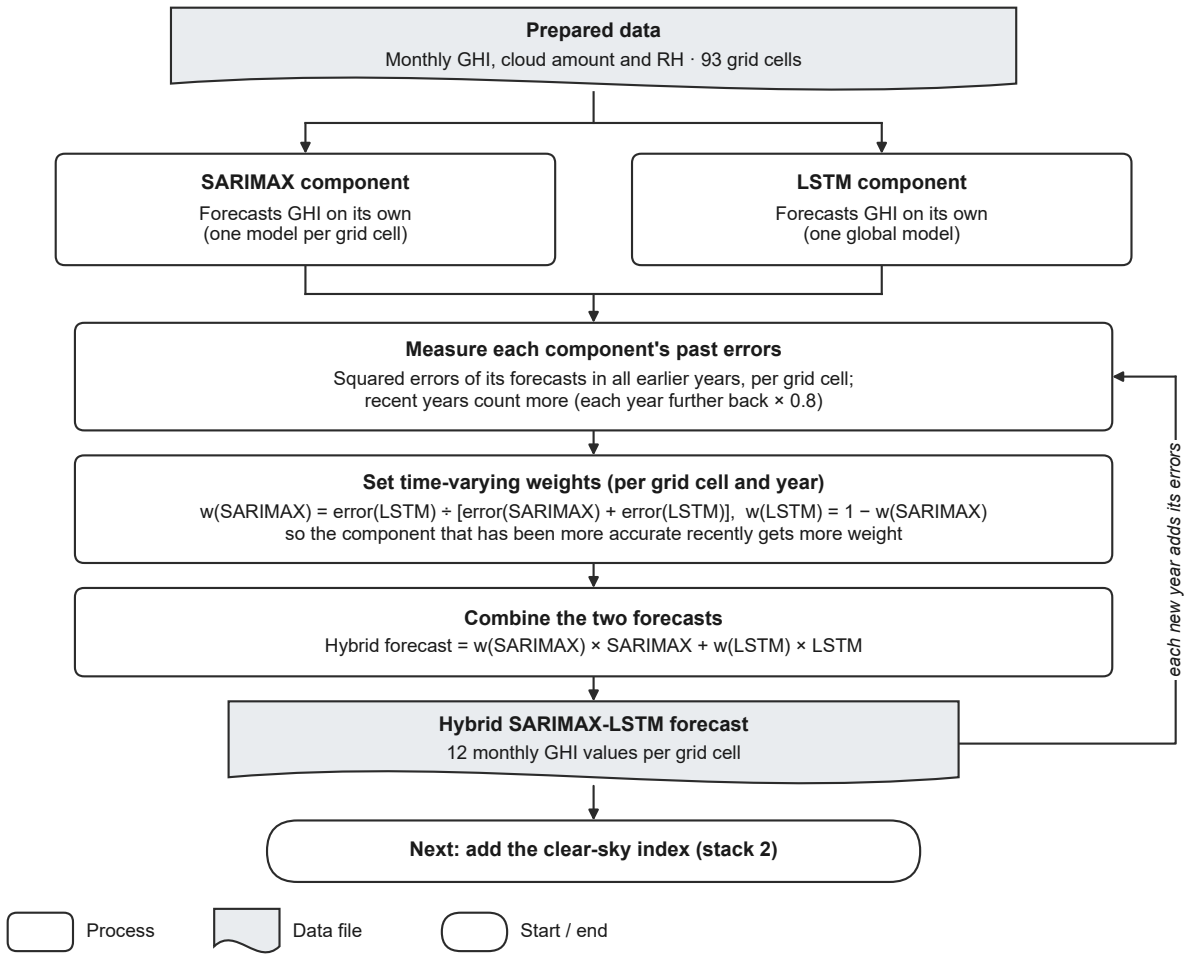

Prepared data
Monthly GHI, cloud amount and RH · 93 grid cells
SARIMAX component
Forecasts GHI on its own
(one model per grid cell)
LSTM component
Forecasts GHI on its own
(one global model)
Measure each component's past errors
Squared errors of its forecasts in all earlier years, per grid cell;
recent years count more (each year further back × 0.8)
Set time-varying weights (per grid cell and year)
w(SARIMAX) = error(LSTM) ÷ [error(SARIMAX) + error(LSTM)], w(LSTM) = 1 − w(SARIMAX)
so the component that has been more accurate recently gets more weight
Combine the two forecasts
Hybrid forecast = w(SARIMAX) × SARIMAX + w(LSTM) × LSTM
Hybrid SARIMAX-LSTM forecast
12 monthly GHI values per grid cell
each new year adds its errors
Next: add the clear-sky index (stack 2)
Process
Data file
Start / end